

Kubernetes Assignment-2

Kubernetes ReplicaSets

Step 1:

1.1 minikube start

minikube start is the command used to create and start a local Kubernetes cluster on your computer using Minikube. It's typically the first command you run after installing Minikube.

1.2 minikube status

minikube status is a command that shows the current state of your Minikube cluster. It tells you whether the cluster and its key components are running.

1.3 minikube start --driver=docker

starts a local Kubernetes cluster using Docker as the virtualization driver instead of creating a separate virtual machine.

1.4 kubectl get nodes

Command is used to list all the nodes in a Kubernetes cluster.

1.5 kubectl get pods -n kube-system

Command is used to list all the pods running in the kube-system namespace, which contains Kubernetes' internal system components.

1.5 kubectl get pods

Command is used to list all the pods in the current namespace (by default, the default namespace).

```
Administrator: Windows PowerShell
Windows PowerShell
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PS C:\WINDOWS\system32> minikube status
3720 status.go:178] status error: host: load: filestore "minikube": open C:\Users\Acer\.minikube\machines\minikube\config.json: The system cannot find the file specified.
3720 status.go:127] status error: host: load: filestore "minikube": open C:\Users\Acer\.minikube\machines\minikube\config.json: The system cannot find the file specified.
PS C:\WINDOWS\system32> minikube delete
* Deleting "minikube" in docker ...
* Removing C:\Users\Acer\.minikube\machines\minikube ...
* Removed all traces of the "minikube" cluster.
PS C:\WINDOWS\system32> minikube start --driver=docker
* minikube v1.38.1 on Microsoft Windows 11 Home Single Language 25H2
* Using the docker driver based on user configuration
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Creating docker container (CPUs=2, Memory=3072MB) ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS C:\WINDOWS\system32> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
PS C:\WINDOWS\system32> kubectl get nodes
NAME                STATUS    ROLES    AGE   VERSION
minikube             Ready     control-plane   2m2s   v1.35.1
PS C:\WINDOWS\system32> kubectl get pods -n kube-system
NAME                READY    STATUS    RESTARTS   AGE
coredns-7d764666f9-1zq7r    1/1      Running   0           2m28s
```

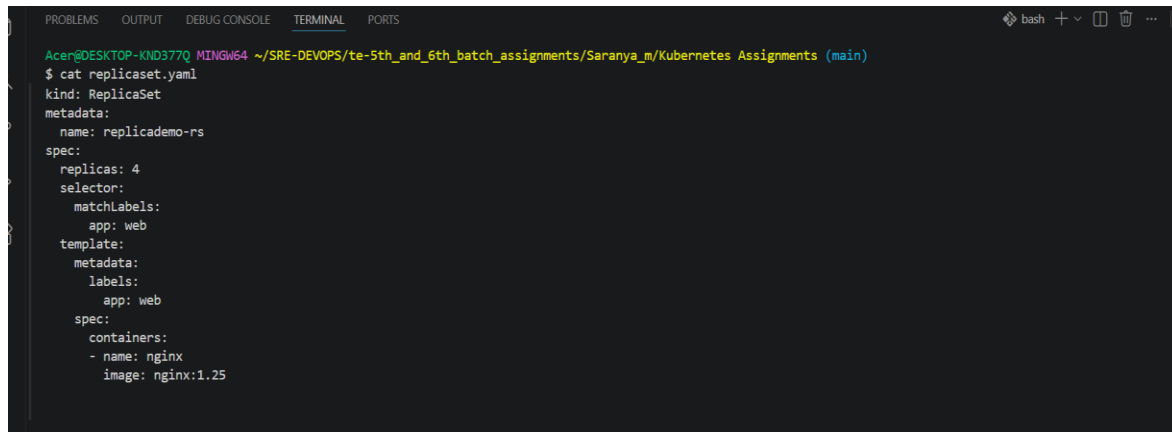
Creating a kubernetes replica set

Step 2: Create a replicaset YAML File

The **vim replicaset.yaml** command is used to create a new YAML file or open an existing YAML

Vim text editor used for editing file.

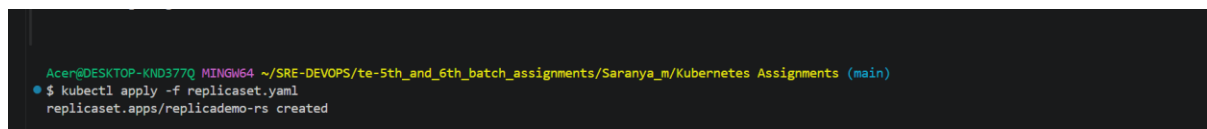
The **cat replicaset.yaml** command is used to display the contents of the **replicaset.yaml** file



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ cat replicaset.yaml
kind: ReplicaSet
metadata:
  name: replicademo-rs
spec:
  replicas: 4
  selector:
    matchLabels:
      app: web
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
        - name: nginx
          image: nginx:1.25
```

Step 3: kubectl apply -f replicaset.yaml

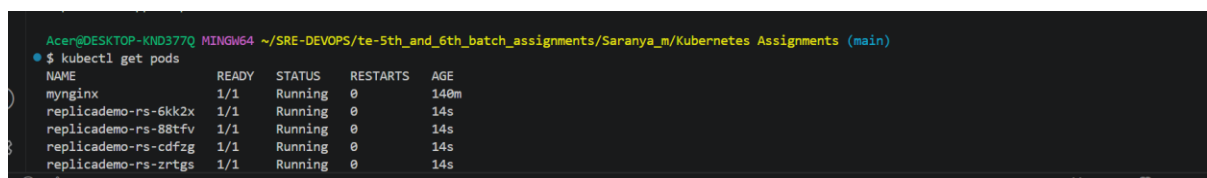
Command is used to create or update a ReplicaSet (or any Kubernetes resource defined in the YAML file) in your Kubernetes cluster.



```
Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl apply -f replicaset.yaml
replicaset.apps/replicademo-rs created
```

Step 4 : Verifying the pod

Command is used to display all Pods in the current namespace.



```
Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
mynginx             1/1     Running   0           140m
replicademo-rs-6kk2x 1/1     Running   0           14s
replicademo-rs-88tfv 1/1     Running   0           14s
replicademo-rs-cdfzg 1/1     Running   0           14s
replicademo-rs-zrtgs 1/1     Running   0           14s
```

Step 5: Verifying the replica set

Kubectl get rs Command is used to list all ReplicaSets in the current namespace.

kubectl describe rs <replicaset-name> Command is used to display detailed information about the ReplicaSet.

```
Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl get rs
replicademo-rs 4 4 4 24s

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl describe rs replicademo-rs
Name: replicademo-rs
Namespace: default
Selector: app=web
Labels: <none>
Annotations: <none>
Replicas: 4 current / 4 desired
Pods Status: 4 Running / 0 Waiting / 0 Succeeded / 0 Failed
Pod Template:
  Labels: app=web
  Containers:
    nginx:
      Image: nginx:1.25
      Port: <none>
      Host Port: <none>
      Environment: <none>
      Mounts: <none>
      Volumes: <none>
      Node-Selectors: <none>
      Tolerations: <none>
Events:
  Type Reason Age From Message
  ----
Normal SuccessfulCreate 2m47s replicaset-controller Created pod: replicademo-rs-6kk2x
Normal SuccessfulCreate 2m47s replicaset-controller Created pod: replicademo-rs-cdfzg
Normal SuccessfulCreate 2m47s replicaset-controller Created pod: replicademo-rs-zrtgs
Normal SuccessfulCreate 2m47s replicaset-controller Created pod: replicademo-rs-88tfv
```

Step 6: Describing a pod

kubectl describe pod <resource-name> Command is used to display detailed information about the displaying detailed information about a specific resource.

```
Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl describe pod replicademo-rs-6kk2x
Name: replicademo-rs-6kk2x
Namespace: default
Priority: 0
Service Account: default
Node: minikube/192.168.49.2
Start Time: Mon, 03 Aug 2026 19:21:25 +0530
Labels: app=web
Annotations: <none>
Status: Running
IP: 10.244.0.13
IPs:
  IP: 10.244.0.13
Controlled By: ReplicaSet/replicademo-rs
Containers:
  nginx:
    Container ID: docker://86889e17ef4b200f3b1bfdce89b3bd0cb190a1612f85a1a48e3178155f6839db
    Image: nginx:1.25
    Image ID: docker-pullable://nginx@sha256:a484819eb60211f5299034ac80f6a681b06f89e55866ce91f356ed7c72af059c
    Port: <none>
    Host Port: <none>
    State: Running
      Started: Mon, 03 Aug 2026 19:21:31 +0530
    Ready: True
    Restart Count: 0
    Environment: <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-nnd2q (ro)
Conditions:
  Type Status
  PodReadyToStartContainers True
  Initialized True
```

Step 7: The **kubectl delete pod <pod-name>** command is used to delete a Pod from a Kubernetes cluster.

The **kubectl delete rs <replicaset-name>** command delete a ReplicaSet in Kubernetes.

```
Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl get pods
NAME READY STATUS RESTARTS AGE
mynginx 1/1 Running 0 144m
replicademo-rs-6kk2x 1/1 Running 0 4m5s
replicademo-rs-88tfv 1/1 Running 0 4m5s
replicademo-rs-cdfzg 1/1 Running 0 4m5s
replicademo-rs-zrtgs 1/1 Running 0 4m5s

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl delete pod replicademo-rs-6kk2x
pod "replicademo-rs-6kk2x" deleted from default namespace

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl get pods
NAME READY STATUS RESTARTS AGE
mynginx 1/1 Running 0 145m
replicademo-rs-88tfv 1/1 Running 0 4m26s
replicademo-rs-cdfzg 1/1 Running 0 4m26s
replicademo-rs-pzw2n 1/1 Running 0 5s
replicademo-rs-zrtgs 1/1 Running 0 4m26s
```

Step 8: Change the image version from nginx 1.25 to nginx 1.30 then check details of pod image version is not changed still it is nginx 1.25. Replicaset does not perform rolling updates.

8.1: First should delete the replicaset and edit the version in yaml file.

8.2: Next run the apply command **kubectl apply -f replicaset.yaml**

8.3: Check the details of pod with describe command there it shows the updated image version from nginx 1.25 to nginx 1.30.

```
Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ cat replicaset.yaml
image: nginx:1.25

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl describe pod replicademo-rs-88tfv
Name:          replicademo-rs-88tfv
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Mon, 03 Aug 2026 19:21:25 +0530
Labels:        app=web
Annotations:   <none>
Status:        Running
IP:            10.244.0.11
IPs:
  IP:          10.244.0.11
Controlled By: ReplicaSet/replicademo-rs
Containers:
  nginx:
    Container ID:  docker://681c22b45b52e5d69c3f0029b3164bf10af8ff2ce58e6ac46b0155410254c2aa
    Image:         nginx:1.25
    Image ID:      docker-pullable://nginx@sha256:a484819eb60211f5299034ac80f6a681b06f89e65866ce91f356ed7c72af059c
    Port:         <none>
```

```
Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl describe pod replicademo-rs-88tfv

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl delete replicaset replicademo-rs
replicaset.apps "replicademo-rs" deleted from default namespace

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl get rs
No resources found in default namespace.

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
mynginx   1/1     Running   0           153m

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl apply -f replicaset.yaml
replicaset.apps/replicademo-rs created

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl get rs
NAME          DESIRED   CURRENT   READY   AGE
replicademo-rs 4         4         0       5s

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl get pods
NAME                  READY   STATUS             RESTARTS   AGE
mynginx               1/1     Running            0           154m
replicademo-rs-47zdt  0/1     ContainerCreating  0           9s
replicademo-rs-98pmj  0/1     ContainerCreating  0           9s
replicademo-rs-k92rb  0/1     ContainerCreating  0           9s
replicademo-rs-nzvvg  0/1     ContainerCreating  0           9s
```

```
Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments (main)
$ kubectl describe pod replicademo-rs-98pmj
Name: replicademo-rs-98pmj
Namespace: default
Priority: 0
Service Account: default
Node: minikube/192.168.49.2
Start Time: Mon, 03 Aug 2026 19:34:39 +0530
Labels: app=web
Annotations: <none>
Status: Running
IP: 10.244.0.18
IPs:
  IP: 10.244.0.18
Controlled By: ReplicaSet/replicademo-rs
Containers:
  nginx:
    Container ID: docker://43157aaf359e036923dc1ad2239e5e75bac809bb3354cd406eba361625ec5cb8
    Image: nginx:1.30
    Image ID: docker-pullable://nginx@sha256:5cf90903deda2c5981b8ad05e7617ac010e389f0dde0ac83487c02c509281de6
    Port: <none>
    Host Port: <none>
    State: Running
      Started: Mon, 03 Aug 2026 19:35:00 +0530
    Ready: True
    Restart Count: 0
    Environment: <none>
```

CrashLoopBackOff:

CrashLoopBackOff is a Kubernetes pod status indicating that a container starts, crashes, and Kubernetes keeps trying to restart it. After each failure, Kubernetes waits progressively longer before each restart, which is called BackOff.

Common causes include:

- Application crashes (exceptions, segmentation faults, bad startup command)
- Missing or incorrect environment variables
- Configuration errors (ConfigMaps/Secrets)
- Liveness or startup probe failures
- Out-of-memory (OOMKilled)
- Missing dependencies (database, Redis, external services)
- Permission or filesystem issues
- Incorrect container entrypoint or command

Quick troubleshooting checklist

- `kubectl get pods` – Confirm the Pod is in CrashLoopBackOff.
- `kubectl describe pod <pod-name>` – Check events, exit codes, and termination reasons.
- `kubectl logs <pod-name> --previous` – Review logs from the crashed container.
- Verify image, command, and environment variables.
- Check ConfigMaps and Secrets.
- Look for OOMKilled or resource limit issues.
- Verify liveness/readiness probes.
- Confirm external dependencies (database, cache, APIs) are reachable.
- Fix the root cause, then redeploy or let Kubernetes restart the Pod.

